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# EE/RPF

## For use from May/November 2027



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Student personal code:

# Extended essay - Reflection and progress form

***Student:*** *This form is to be completed by you and your supervisor during the progress and completion of the extended essay. It records your progress through the extended essay process and includes your reflective statement addressing your extended essay experience. You must undertake three formal reflection sessions with your supervisor: The first formal reflection session should focus on your initial ideas and how you plan to undertake your research; the interim reflection session takes place once a significant amount of your research has been completed, and the final session will be in the form of a viva voce, once you have completed and handed in your extended essay. This document acts as a record in supporting the authenticity of your work. Following the viva voce, you write a* ***reflective statement*** *of no more than 500 words., which must be included on the RPF.*

**The completion of this form is a mandatory requirement of the extended essay. It must be submitted together with the completed essay. The reflective statement will then be assessed under Criterion E. Students and supervisors should be aware that a mark of 0 will be awarded by the examiner for criterion E if the RPF is blank or the reflective statement is written in a language other than that of the accompanying essay.**

***Supervisor:*** *You must have three reflection sessions with each student, one early in the process, an interim meeting and then the final viva voce. Other check-in sessions are permitted but do not need to be recorded on this sheet. After each reflection session you must record the month that the reflection session took place, the student's year of DP study at that time and initial the form.*
***By submitting this student work for assessment, you are taking responsibility for its authenticity. No piece of student work should be uploaded/submitted to the e-Coursework system if its authenticity is in doubt.***

***First reflection session***

Month: March DP year (1 or 2): 1 Supervisor initials:

***Interim reflection session***

Month: November DP year (1 or 2): 1 Supervisor initials:

***Final reflection session – Viva voce***

Month: January DP year (1 or 2): 2 Supervisor initials:

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## Reflective statement

(No more than 500 words, written in the same language as the accompanying extended essay)

I've gained interest in the topic of sustainable packaging through my design technology course. I was fascinated by the innovative and creative ways environmental issues are approached. I live in a time and place where these innovations are implicated in our daily lives. My interest was further developed by visualizing and experiencing the process of changes in society where an issue is recognized, communities are formed to raise awareness, and new solutions/alternatives are implicated over time. Some examples are plastic straws being replaced by paper straws, plastic bags being charged, or replaced by paper bags. As this topic is relevant locally as it is globally, it allows me to get into more depth by narrowing down my target audience as I can be more engaged with users through primary research rather than putting more weight on secondary research.

I will to make a decision wether I will be conducting this essay on a local or a global basis. The I will choose in specific what sector of plastic food packaging I will be looking into; possible options are traditional plastic bags, per bottles, PVC containers, styrofoam. The next step is to conduct a secondary research of this type of packing material such as production statistics, possible waste management, and lids cycle. This will set the base of researching the issues regarding the use of specific material in food packaging. I will also gain knowledge about the application of this material and what material properties were essential in the application which allows me to construct a research for possible solutions such as manufacturing methods or use of more environmentally friendly material as an alternative which fulfils the specific material properties.

To propose the question "The impact mass manufactured food packaging has on the plastic waste generated in the world", I will need information on the production of single use food packaging and how it has ended up becoming plastic waste. Then I will be carrying out quantitative data to visualize the production, consumption, emissions, expots, environmental impacts. Field research will be consulted in a local community based survey towards consumers of food packaging waste. As food packaging is a common spending to most communities and individuals, therefore the data will be relative when collected from surveys within a local community based such as schools the content of the survey will mainly focus in client cost constraints and possible vision to improve environmental impact of single use container.

I've started my research by looking into the market application of different food packaging. One complication I faced at the beginning of my research was that this was already a solved issue for more large restaurants.

According to my research, most large franchise restaurants are under strict regulations on the use of plastic food packaging in recent years; replacing most plastics with paper. However, local independent restaurants have not been as heavily affected by this regulation; and through primary research, I was able to find out that most of these restaurants still depend on plastic packaging.

*** Note: 502 words up to this point ***

In addition, my survey showed that more than half of the students stated that they order more from local independent restaurants than large chains. Therefore, I've focused on packaging of take-aways and deiveries in local independent restaurants. For my secondary research, I've looked into the market, application, properties/characteristics, and environmental issues of polystyrene foam and polypropylene.

From experiences of ordering food or using takeaways services, I've started to recognize some restaurants using plastic containers. By reaching out to some of these shops nearby; I've conduct primary research to gain information about the sales, supply, cost and, application. to gain information from the consumer viewpoint; I've run a survey to students from school; asking questions regarding consumption, values, preference, etc of consumers.

I've repeated this research method for other substitutional packaging materials such as PLA-coated kraft paper and sugarcane fiber which was important for comparing and evaluating different options. To write an in-depth essay and present accurate and convincing arguments, I've looked into the life cycle analysis; gaining information about the raw materials, transportation of raw materials, production, distribution on the marker, and end life of the materials.

This research was especially challenging as the information I was trying to find was difficult or unclear due to it being inconsistent by region.

Final reflection conducted with student. Student has a realistic opinion of their work. Work improved from initial draft and student was able to relate the EE to their past and future plans.

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